“Understanding“ as meant by biologists or by physicists: the example of vegetal stems growth

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For several centuries the growth of plants has been investigated by biologists, physicists and mathematicians. However their approaches have remained distinct because what is meant by “understanding“ is different in these different scientific communities. Far from suggesting that a unifying view-point should be reached, I will argue that this situation has permitted the interplay of several different research strategies that have been particularly fruitful not only for the progress in a particular problem but also for both biology and physics in general. I will discuss this issue by revisiting several works concerning the apical growth and the phyllotaxis (the organization of e.g. leaves around the stem) that are characteristic of most land plants. For the physicist vegetal growth is one amongst many morphogenetic phenomena in living and non-living systems in which he seeks universal features of self-organization. By contrast the biologists are more interested by the physiological processes involved in this growth, by the genetic control pathways and by the evolutionary process by which this type of growth has come to dominate. As an example of the complementarity of the biological and physical approaches I will describe a recent work on the morphology of a brown alga *Sargassum muticum* that has a global structure comparable to that of a land plant, including a Fibonacci related “phyllotaxis“. This is remarkable since it cannot result from a common inheritance: brown algae and land plants are two clades that diverged approximately 1800 million years ago when they were both unicellular organisms. The observed similarity in morphology is thus a convergence. As seen by the biologists it is the result of a similar selective pressure, for the physicist it comes in support of the genericity of the self-organization processes. These two viewpoints are compatible and complementary; they give a perspective on how evolution “uses“ and implements physical processes in the build-up of living entities.